Yajesh Chandra

Email: yajeshstan@gmail.com
LinkedIn | GitHub | Codeforces
Mobile: +91 7289841708

Education

Indian Institute of Technology, Mandi — Himachal Pradesh, India
B.Tech in Computer Science and Engineering — *Aug 2023 – Current*

- **Relevant Coursework:** Machine Learning, Deep Learning, Data Structures and Algorithms, Design of Algorithms, Information and Database Systems, Mathematical Foundations of Computer Science, Formal Languages and Automata Theory, Computer Organization, Probability & Statistics

RWTH Aachen University — Aachen, Germany
Exchange Semester — *Oct 2025 – Feb 2026*

- **Relevant Coursework:** Computer Graphics, Artificial Intelligence, Virtual Reality, Quantum Computing

Experience

Tech Mahindra — GitHub
Software Intern | Tempest – Local Security Analysis POC — *May 2026 – Jul 2026*

- **Solution Architecture:** Translated an open-ended security problem into a Python static analysis engine to detect non-production code accessing production resources and violating read/write access policies.
- **Graph Analysis:** Modeled function calls, imports, and assignments through AST-derived code graphs; implemented NetworkX DiGraph annotations and YAML policy rules, handling **200k+ graph nodes**.
- **Privacy by Design:** Selected fully local analysis over cloud processing to avoid exposing proprietary source code; integrated the engine with AI developer tools through the **Model Context Protocol (MCP)**.
- **Incremental Processing:** Implemented Git-hash-based incremental synchronization to update changed code and retain unchanged graph data, avoiding unnecessary full-repository recomputation.

Projects

IndexIt – AI-Powered PDF Indexing Platform — GitHub
Next.js, TypeScript, Python, FastAPI, PaddleOCR, PyTorch — *Sept. 2026*

- **Full-Stack Architecture:** Built a PDF indexing platform using **Next.js and FastAPI**, integrating OCR, NLP-based topic segmentation, and contextual heading generation into an end-to-end document processing pipeline.
- **Large-File Processing:** Implemented **5 MB chunked uploads** supporting PDFs up to **100 MB**, with upload progress tracking, multithreaded OCR processing, and direct text extraction for digital documents.
- **API Development:** Developed **REST APIs** for weighted multi-region content extraction, contextual heading regeneration, and document editing; enabled PDF exports with clickable bookmarks and hierarchical navigation.

MatBot – RAG-Based MATLAB Troubleshooting Assistant — GitHub
Python, LangChain, Mistral-7B, ChromaDB, HuggingFace, Streamlit — *May 2025*

- **LLM & RAG Engineering:** Developed an **LLM-powered MATLAB troubleshooting assistant** using **LangChain and Mistral-7B**, implementing Retrieval-Augmented Generation (RAG) to retrieve technical documentation and generate context-aware debugging solutions.
- **Semantic Search & Vector Database:** Used **ChromaDB** and **BAAI/bge embeddings** for semantic search; ranked documentation by cosine similarity to ground generated debugging responses.
- **API Integration & User Experience:** Integrated **Tavily Search API and Wikipedia** for external retrieval; built a **Streamlit chat interface** supporting text queries and MATLAB error screenshots.

Technical Skills

Programming: C/C++, Python, TypeScript, JavaScript, SQL, Bash
Backend & Web: FastAPI, REST APIs, Next.js, React, Streamlit, HTML/CSS
AI & Data: PyTorch, scikit-learn, Transformers, LangChain, RAG, ChromaDB, NumPy, Pandas
Systems & Tools: Linux, Git/GitHub, Docker, NetworkX, MCP, ROS/ROS2, CMake

Achievements

- **1st place, HCL Tech Hackathon** (Deep Learning course), among **50+ UG/PG teams**; **Rs. 50,000** prize.
- **1st place, Robotics, FrostHack 2025** (inter-college), for a CV-based localization system; **Rs. 30,000** prize.
- **JEE Advanced: AIR 2909** and **JEE Mains: 99.70 percentile** (2023; 1M+ candidates).